Check for updates

MANY-BODY DYNAMICS

Observation of many-body dynamical localization

Yanliang Guo[1]†, Sudipta Dhar[1]†, Ang Yang[2]†, Zekai Chen[1], Hepeng Yao[3], Milena Horvath[1], Lei Ying[2]*, Manuele Landini[1], Hanns-Christoph Nägerl[1]*

The quantum kicked rotor is a paradigmatic model system in quantum physics. As a driven quantum system, it features dynamical localization, specifically Anderson localization in momentum space. However, the interacting many-body kicked rotor is believed to break localization. Here, we present evidence for many-body dynamical localization for the Lieb-Liniger version of the many-body quantum kicked rotor. After some initial evolution, the momentum distribution of interacting quantum-degenerate bosonic atoms in one-dimensional geometry, kicked hundreds of times by means of a pulsed sinusoidal potential, stops spreading. Our results shed light on the boundary between the classical, chaotic world and the realm of quantum physics.

Chaos is a phenomenon that is found almost everywhere in our daily life, and it plays a central role in many of the sciences. Nonlinear complex systems, which appear abundantly in mathematics, physics, biology, ecology, meteorology, economics, and other fields, are generally subject to chaotic dynamics (*1*). The long-time evolution sensitively depends on the initial conditions and is inherently unpredictable. Chaos, although it has a negative connotation, is often very useful. It is intimately connected to ergodicity, and dense sampling of the available phase space drives many important processes (*2*), from thermodynamics to biology. Although the emergence of chaos in classical systems is well understood (*2*), its appearance in quantum physics and its relation to relaxation and thermalization in quantum many-body systems have been a topic of ongoing debate since the very beginnings of quantum physics (*3*, *4*). Most quantum systems do relax, though more and more systems have been identified by theory and in experiment for which ergodicity is evidently broken. Examples include many-body localized cold atoms in lattices (*5*), many-body scar states of interacting Rydberg atom arrays (*6*), and prethermalized samples of bosons in low dimensions (*7*, *8*). Central questions for quantum many-body physics are hence: What ingredients are needed to restore ergodicity? When does a drive, i.e., the pumping of energy into the system, lead to chaos and subsequently to heating? Do driven interacting many-body systems necessarily have to thermalize?

A minimal physical system to study chaotic behavior in a driven system is a rotating object periodically kicked by an external force (*9*). A transition from integrable to chaotic diffusive motion is found as the kick strength crosses a critical value. Its quantum version, known as the quantum kicked rotor (QKR) (*9*, *10*), is a paradigmatic example in which quantum coherence can prevent the system from absorbing energy even in the strong-driving regime, in stark contrast to the classical counterpart. This quantum interference–induced phenomenon can be understood as Anderson localization in momentum space (*11*–*13*). It is termed dynamical localization (DL) and has been demonstrated experimentally in dilute ultracold atomic gases (*14*–*16*).

The QKR is a highly idealized single-particle model system. Realistic systems consist of many particles that interact with each other. Interactions almost always lead to a randomization of the many-body dynamics (*17*, *18*). Driving an interacting quantum many-body system further increases the complexity. For the case of the many-body QKR, a breakdown of DL has been proposed using mean-field approaches (*19*–*21*), and indeed, recent cold-atom experiments with either weakly interacting one-dimensional (1D) (*22*) or strongly interacting 3D gases (*23*) align with these proposals. However, recent theoretical work (*24*–*28*) suggests that DL may persist in an interacting, even strongly interacting, many-body quantum system. Specifically, interacting 1D bosonic gas as an ideal Lieb-Lininger QKR is proposed to support many-body dynamical localization (MBDL). Driving a strongly correlated interacting system surprisingly may not have to result in chaotic dynamics or in a thermalized state.

Here we report the observation of MBDL in interaction-tunable 1D Bose gases. Using quantum-degenerate samples of Cs atoms, we realize the Lieb-Liniger QKR, a paradigmatic model for the simulation of driven interacting quantum many-body systems. We tune the interaction strength from the noninteracting regime to the Tonks-Girardeau (TG) regime (*29*–*32*) and find signatures of MBDL in the freezing of the momentum distribution and in the saturation of the energy and the entropy as the 1D Bose gases are kicked hundreds of times. The first-order correlation function exhibits a markedly different decay in the two limiting cases of zero and strong interactions. Our results are in agreement with recent proposals (*24*–*26*) and are captured qualitatively by our modeling.

Experimental setup and model

The experiment starts by loading a Bose-Einstein condensate (BEC) of Cs atoms (*33*) into an array of narrow 1D tubes created by a 2D optical lattice as illustrated in Fig. 1A. The laser beams forming the lattice propagate in the horizontal *x*-*y* plane at right angle to each other, and their power is adiabatically ramped up to give a lattice depth of $30E_r$, where $E_r = \pi^2\hbar^2/(2ma^2)$ is the recoil energy. Here, $a = \lambda/2 = \pi/k_L = 532.25$ nm is the lattice spacing with the wave number k_L that is set by the wavelength λ of the lattice light, and m is the atomic mass. At this depth, tunneling between the tubes is fully suppressed, and the tubes can be considered as independent 1D systems (*34*, *35*). A magnetic field B along the vertical z-direction controls the interatomic interaction (*36*) as quantified by γ, the dimensionless 1D Lieb-Liniger interaction parameter (*37*). By choosing values between $\gamma = 0$ and $\gamma = 11$, we cover the range from the noninteracting, single-particle regime well into the strongly interacting TG regime in which the bosons have fermionized. The atoms are levitated against gravity by means of a gradient of B. The finite transversal extent of the lattice beams results in weak harmonic trapping with trap frequency $\nu_z = 14.7(3)$ Hz along the tubes' direction. We partially compensate this trap by means of a horizontally propagating blue-detuned antitrap laser beam. This forms a flat-bottom potential, as illustrated in Fig. 1B. Typically, we fill about 10^4 tubes, with a weighted average of 18 atoms per tube. At this stage, the 1D systems are in equilibrium with a temperature of ~2 nK, which is estimated by the thermometry method demonstrated in (*38*, *39*). Calculated 1D density distributions $n_{1D}(z)$ for various values of γ are included in Fig. 1B.

We next implement the QKR procedure. A longitudinal standing wave with lattice spacing a and depth V_z is periodically pulsed on with period T and square-pulse duration T_p for a number of N_p pulses as illustrated in Fig. 1C. We choose $T = 60$ μs and 80 μs, $T_p = 10$ μs, and $V_z = 20E_r$, resulting in a dimensionless kick strength $K = 3.3$ and $K = 4.4$ respectively; see below and (*40*) for the definition of K. For values of N_p between 1 and 1101, the entire pulse sequence takes up to 66 and 88 ms, respectively. Figure 1D illustrates the density distribution for the many-body case during the kicking procedure. We obtain the longitudinal momentum distribution $n(k)$ using a 20-ms time-of-flight (TOF) absorption-imaging technique. A typical absorption image is shown in Fig. 1E, and $n(k)$ is obtained as a probability distribution by integration along the transversal direction and normalizing by the atom number. For a faithful

[1]Institut für Experimentalphysik und Zentrum für Quantenphysik, Universität Innsbruck, Innsbruck, Austria. [2]School of Physics and Zhejiang Key Laboratory of Micro-nano Quantum Chips and Quantum Control, Zhejiang University, Hangzhou, China. [3]DQMP, University of Geneva, Geneva, Switzerland. ***Corresponding author. Email: leiying@zju.edu.cn (L.Y.); christoph.naegerl@uibk.ac.at (H.-C.N.)** †These authors contributed equally to this work.

Downloaded from https://www.science.org at Hong Kong Polytechnic University on March 16, 2026

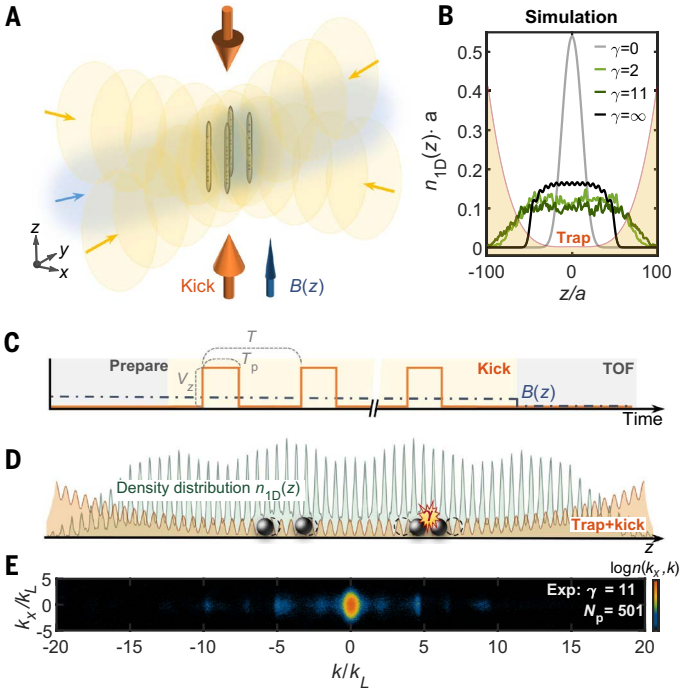

Fig. 1. Experimental implementation of MBDL. (**A**) Sketch of the 3D BEC loaded into an array of 1D tubes generated by the 2D optical lattice (yellow) and partially compensated by a blue-detuned optical anti-trap (blue). The QKR lattice propagates along the vertical z direction. (**B**) Calculated ground-state many-body 1D spatial density $n_{1D}(z)$ for the flat-bottom trap in units of a^{-1} for various values of the interaction parameter γ. The data for $\gamma = 2$ and $\gamma = 11$ were obtained from a quantum Monte Carlo calculation at a temperature of 2 nK, and the data for $\gamma = 0$ and $\gamma = \infty$ are the zero-temperature results from a one-cycle Floquet map and fermionization method (*32*,*40*,*42*). (**C**) Schematic of the experimental QKR sequence with N_p pulses sandwiched between the preparation and TOF-imaging procedures. (**D**) Illustration of $n_{1D}(z)$ (green) for four interacting atoms in a cosine standing-wave potential in the flat-bottom trap (orange) for $\gamma = \infty$. The anti-nodes of $n_{1D}(z)$ register with the nodes of the kick potential. The envelope of $n_{1D}(z)$ with the four bumps reflects the presence of four atoms. (**E**) Typical absorption image for $\gamma = 11$ after $N_p = 501$ kicks showing the spatial density after TOF in color coding (the detailed amplitude can be seen in the momentum distribution of Fig. 2) and integrated along the line-of-sight.

measurement of $n(k)$, the particle interaction is switched off via a Feshbach resonance's zero crossing (*36*) during TOF. From $n(k)$, we subsequently compute the kinetic energy E, the information entropy S, and the one-body correlation function $G^{(1)}$.

We model the system by the explicitly time-dependent Lieb-Liniger QKR Hamiltonian. For N interacting bosons with mass m moving in a 1D tube at zero temperature, it reads (*24*, *26*)

$$H(t) = \sum_{i=1}^{N}\left[\frac{P_i^2}{2m} + \hbar\kappa\cos\left(2k_L z_i\right)\sum_{N_p}\delta_{T_p}\left(t - N_p T\right) + V_{ext}\left(z_i\right)\right] + g_{1D}\sum_{i<j}^{N}\delta\left(z_i - z_j\right) \quad (1)$$

Here, $\hbar\kappa = V_z T_p / 2$ sets the kick strength $K = 8TE_r\kappa/\hbar$. The first three terms represent the QKR for a single particle at position z_i moving in the trap potential V_{ext}. The last summand models the δ-type pairwise contact interaction whose strength g_{1D} is related to γ via $\gamma = mg_{1D}/n_{1D}\hbar^2$. Details of our modeling are given in (*40*).

Freezing of the momentum distribution

We first turn to the evolution of the momentum distribution as N_p is increased. Figure 2 presents the experimental results of $n(k)$ for two limiting values of the interaction strength, namely for vanishing ($\gamma = 0$) and for very strong interactions ($\gamma = 11$). In the former case, the system is in the single-particle regime; in the latter, it is deeply in the TG regime. Starting with the first kick, higher-momentum states are populated, primarily the ones with $\pm 2\hbar k_L$. For the noninteracting case, some of the population is subsequently transferred to higher-momentum states, but eventually further transfer is halted and the distribution becomes stationary, as can be seen from comparing the data sets with $N_p = 301$ and $N_p = 1001$. The system is dynamically localized (*15*, *16*, *23*). In the interacting case, the change in the momentum distribution from $N_p = 1$ to about $N_p = 301$ is much more pronounced. However, the subsequent evolution, in particular from $N_p = 501$ to $N_p = 1101$, appears to have stopped. Examples of $n(k)$ after the first kick and in the localized state after $N_p = 801$ for the noninteracting and the strongly interacting cases are shown in Fig. 2, C and D, respectively. As we will argue below, the interacting system, after some initial evolution toward a broadened distribution, has entered the regime of MBDL.

We use the Jensen-Shannon divergence J to quantitatively study the evolution of $n(k)$. It is a measure of the overlap of two probability distributions (*41*) with values between $J = 0$ (full overlap) and $J = 1$ (no overlap) (*40*). Figure 3A presents the time evolution of J between the momentum distribution at N_p and $N_p + 100$ for both the noninteracting and the strongly interacting cases, as N_p is stepped in units of 50. Initially, as the distributions still change, the value for J is large compared to the noise floor. But quite quickly, J drops toward the noise level. For the noninteracting case, the noise level is reached already after the first step, and overlap cannot be distinguished from noise beyond $N_p = 300$. The system has undergone dynamical localization. For the interacting case, this takes about $N_p = 450$ kicks. Afterwards, the change in the distributions is minimal and can no longer be distinguished from the noise. The system has undergone MBDL. This can also be seen from a direct comparison of two typical measured distributions taken at $N_p = 701$ and $N_p = 1001$ for the strongly interacting case $\gamma = 11$, as shown in the inset of Fig. 3A. Within the error bars, the two structured distributions are the same; even the slight shifts away from the multiples of $2\hbar k_L$ are well aligned. This is notable, as the interacting system has been kicked 300 times more for the later measurement. Evidently, the interplay of quantum coherence and strong interactions prevents the breaking of localization.

Saturation of the energy and entropy

We further quantify the behavior observed above by analyzing the longitudinal kinetic energy $\langle E_z \rangle \propto \langle k^2 \rangle$ and the information entropy $S = -\sum_k n(k)\log[n(k)]$ for various values of γ. The former is measure of the absorbed energy, and the latter relates to the degree of thermalization in the system. The data are taken for two values of the kick strength, $K = 3.3$ ($T = 60$ μs) and $K = 4.4$ ($T = 80$ μs), and, in addition to the previous choice of $\gamma = 0$ and $\gamma = 11$, for intermediate values of γ. The experimental results are shown in Fig. 3, B and D. For all values of γ, both observables, after an initial rise, settle to constant values after a few hundred kicks. Evidently, the evolution has stopped. The limiting values depend on γ: The larger γ, the larger the localization energy and the localization entropy for a given K. The limiting values also depend on K. One can see a clear difference for the data taken for $\gamma = 11$, with a smaller kick strength but a higher kick frequency giving larger localization values. However, all this behavior is in stark contrast to what happens when the system is subject to random kicks. In that case, as shown in Fig. 3B, the energy increases diffusively without bounds. For both the strongly and the noninteracting scenarios, $n(k)$ ultimately ends up as a uniform distribution in our measured momentum window [see details in (*40*)]. Therefore, quantum coherence is crucial to the emergence of the DL and MBDL phases. We note that for a time evolution up to $N_p = 1001$, atom loss is small and transversal

Downloaded from https://www.science.org at Hong Kong Polytechnic University on March 16, 2026

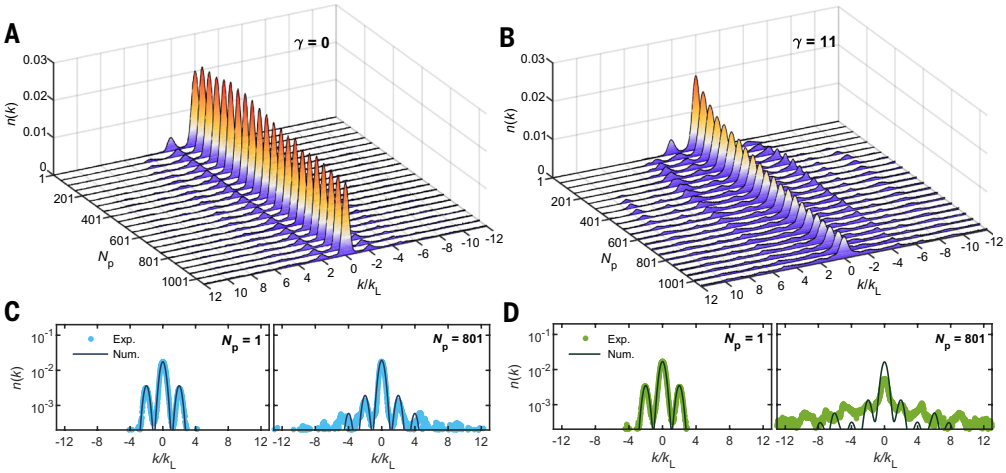

Fig. 2. Evolution of the 1D momentum distribution. Measured $n(k)$ for the noninteracting (**A**) and strongly interacting case (**B**) for a kick strength $K = 3.3$. In (A) and (B), the number of kicks N_p is increased from 1 to 1051 in steps of 50. Example distributions are shown in (**C**) and (**D**) for $N_p = 1$ and 801 for $\gamma = 0$ and $\gamma = 11$, respectively. The data are compared to numerically calculated distributions for $\gamma = 0$ and $\gamma = \infty$ (*40*). Because of the finite TOF duration, these are convoluted by Gaussians whose waist is half of the flat-bottom length. Note also that the higher-order momentum peaks are shifted to higher momentum values by up to $1\hbar k_L$. This also results from the fact that the TOF is finite as the atoms effectively experience some TOF during the evolution before they are released (*23*). All experimental distributions are the average of three realizations.

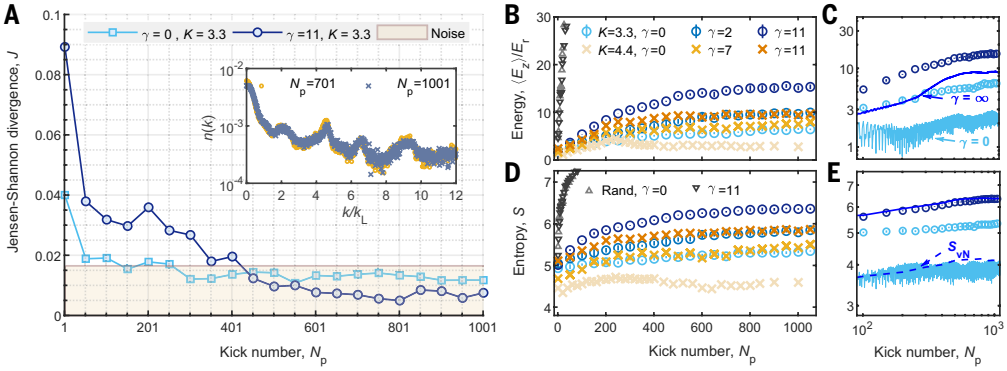

Fig. 3. Evidence for MBDL. (**A**) Jensen-Shannon divergence J between the momentum distributions at N_p and $N_p + 100$ as N_p is stepped in units of 50, for the noninteracting ($\gamma = 0$, light blue squares) and the strongly interacting ($\gamma = 11$, dark blue circles) cases. The orange shaded area indicates the typical shot-to-shot noise determined by comparing three repeats at $N_p = 501$. Inset: Comparison of measured $n(k)$ at $N_p = 701$ and at $N_p = 1001$ for the strongly interacting case $\gamma = 11$. (**B** and **D**) Time evolution of the kinetic energy $\langle E_z \rangle$ (B) and of the information entropy S (D) for various values of the 1D interaction strength γ as indicated and for two values of the kick strength $K = 3.3$ ($T = 60$ μs, circles) and $K = 4.4$ ($T = 80$ μs, crosses). For comparison, the measured time evolution of noninteracting ($\gamma = 0$, gray up-triangles) and strongly interacting ($\gamma = 11$, black down-triangles) gases is shown for random kicks. (**C** and **E**) Comparing the energy (C) and entropy (E) calculated for the noninteracting (light blue solid line) and infinitely interacting cases (dark blue solid line) (*40*) to the data with $\gamma = 0$ (light blue circles) and $\gamma = 11$ (dark blue circles) for the case of $K = 3.3$ on a log-log scale. Fast oscillations for the theory data for $\gamma = 0$ are not resolved. Additionally, the calculated von-Neumann entropy S_{vN} for infinite interactions is shown (dashed line). The statistical error in (B) through (E) is smaller than the symbol size.

excitations are negligible. Stronger interactions lead to slightly higher atom loss, but losses remain below 25%. We also note that we can increase γ to values that are much larger than the ones used here, e.g., to $\gamma = 86$. For this choice, we find increased atom loss and pronounced transversal excitations. These become evident already after about 100 kicks. Details on loss and transversal excitations are discussed in (*40*).

In Fig. 3, C and E, we compare the results for vanishing and very strong interactions to our theory data, obtained in the two limiting cases $\gamma = 0$ and $\gamma \to \infty$, for which calculations can be carried out. Although the experimental data lie above the theory predictions, the growth trend for the energy and the entropy for experiment and theory agree reasonably well. In particular, theory confirms the onset of localization after around 500 kicks. The theory allows us to also calculate the von-Neumann entropy $S_{vN} = -\mathrm{Tr}[\rho \log \rho]$, where ρ is the one-particle density matrix (*40*, *42*, *43*). It is a measure of the entanglement generated in the system. It also settles to a constant value, and its growth trend is almost the same as for the information entropy.

One-body correlation function

We now turn to the quantum coherence of our kicked many-body system. It is contained in the one-particle density matrix $\rho(z, z') = \langle \hat{\Psi}^\dagger(z') \hat{\Psi}(z) \rangle$, where $\hat{\Psi}$ is the many-body field operator. It governs the functional form of the one-body correlation function via $G^{(1)}(z) \propto \int \rho(z, z') dz'$. In practice, one gets $G^{(1)}(z)$ from the Fourier transform of $n(k)$ (*29*, *39*). The experimental results for $G^{(1)}(z)$ of the QKR are presented in Fig. 4 for the TG regime and the noninteracting regime for two kick strengths and for three values of N_p in the localized regime. As a consequence of the frozen momentum distribution, $G^{(1)}(z)$ does not depend on N_p after MBDL has occurred (the data for different values of N_p are offset for visibility). The decay for the strongly interacting case is well fit by an exponential for intermediate distances. A frozen correlation length is considered to be a property of the effectively thermalized TG gas in MBDL phase, as predicted for trapping-free systems (*26*, *27*). The exponential fit is not expected to cover $G^{(1)}$ for very short distances, which via Fourier transform correspond to the region of large momenta. This is compatible with a predicted algebraic-like decay for large momenta in the MBDL phase (*25*–*27*). Our data are in fact consistent with a $n(k) = C_{th} k^{-4}$ behavior. Here, the weight C_{th} is Tan's contact of the effectively thermalized TG gas (*26*, *44*, *45*) [see details in (*40*)]. Contrary to the TG case, for the noninteracting system, $G^{(1)}(z)$ exhibits a Lorentzian decay behavior. This implies an exponential decay in momentum space, signaling Anderson-like localization (*11*). Although the one-body correlation function for the MBDL phase requires more theoretical studies, especially for finite interactions and for trapped systems, the change in the behavior of the correlation function from one regime to the other highlights the many-body effect in the MBDL dynamics.

Downloaded from https://www.science.org at Hong Kong Polytechnic University on March 16, 2026

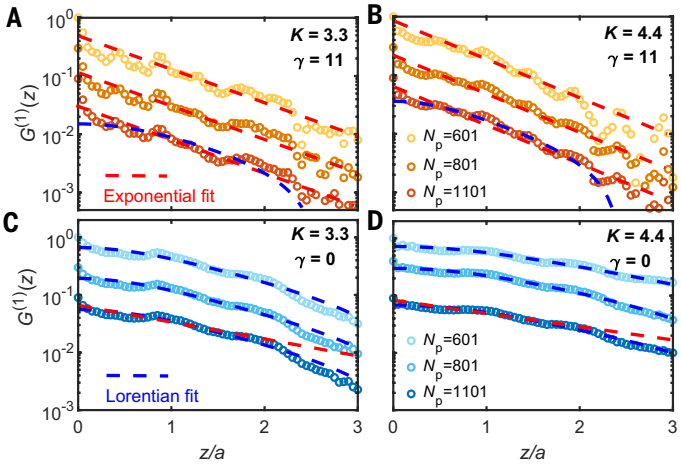

Fig. 4. Decay of the one-body correlation function for DL and MBDL. (**A** and **B**) One-body correlation function $G^{(1)}(z)$ for strong interactions ($\gamma = 11$) and two kick strengths and for $N_p = 601$, 801, and 1101, as indicated. The different datasets have been offset for better visibility. Exponential fits to the data for the range $z/a = 0.25$ to 2.5 are indicated by the red dashed lines. (**C** and **D**) Same as (A) and (B), but for the noninteracting case ($\gamma = 0$). Lorentzian decays have been fit to the experimental data over the full range (blue dashed lines). For comparison, Lorentzian and exponential fits are included in the plots shown in [(A) and (B)] and [(C) and (D)], respectively.

Discussion and outlook

In this study, we have observed and analyzed a many-body phase termed MBDL by periodically kicking a TG gas confined in a flat-bottom trap. Our observations reveal that the momentum distribution of a strongly correlated system freezes despite the periodic kicks and preserves its characteristic interference structure. The kinetic energy and the information entropy show suppressed growth and saturation. We conclude that the quantum many-body system does not fall into the regime of thermalization and chaos. The differences for the momentum distributions and the correlation functions between the noninteracting scenario and the strongly interacting one highlight the impact of the many-body effects under the periodic driving. Our findings raise a series of questions to experiment and theory: As the absence of thermalization does not exclude the presence of chaos (*46*), then what is the role of chaos in the MBDL phase, and how can it be quantified experimentally and by theory? What other mechanisms apart from random kicking could break MBDL in our Lieb-Liniger setting, and how do chaos and thermalization emerge from these? What happens for intermediate interactions in the presence of inhomogeneous trapping for which integrability is expected to be broken? As interactions for our system are short-ranged in real space and hence necessarily long-ranged in momentum space, is there a mapping to real-space localized states for the case of long-range real-space interactions (*47*, *48*)? How is the relation to other many-body localized states that are, e.g., found in a lattice setting (*49*) for which localization happens in real space? The observation of MBDL, in view of recent theoretical research (*24*–*27*), opens the door to a further exploration of the emergence of chaos in a many-body setting.

REFERENCES AND NOTES

1. S. H. Strogatz, *Nonlinear Dynamics and Chaos with Student Solutions Manual: With applications to Physics, Biology, Chemistry, and Engineering* (CRC press, 2018).
2. G. L. Baker, J. P. Gollub, *Chaotic Dynamics: An Introduction* (Cambridge Univ. Press, 1996).
3. M. Berry, *Phys. Scr.* **40**, 335–336 (1989).
4. L. D'Alessio, Y. Kafri, A. Polkovnikov, M. Rigol, *Adv. Phys.* **65**, 239–362 (2016).
5. P. Bordia, H. Lüschen, U. Schneider, M. Knap,I. Bloch, *Nat. Phys.* **13**, 460–464 (2017).
6. H. Bernien *et al.*, *Nature* **551**, 579–584 (2017).
7. Y. Le, Y. Zhang, S. Gopalakrishnan, M. Rigol, D. S. Weiss, *Nature* **618**, 494–499 (2023).
8. D. Abanin, W. De Roeck, W. W. Ho, F. Huveneers, *Commun. Math. Phys.* **354**, 809–827 (2017).
9. F. Haake, *Quantum Signatures of Chaos* (Springer, 1991).
10. M. Santhanam, S. Paul, J. B. Kannan, *Phys. Rep.* **956**, 1–87 (2022).
11. P. W. Anderson, *Phys. Rev.* **109**, 1492–1505 (1958).
12. S. Fishman, D. R. Grempel, R. E. Prange, *Phys. Rev. Lett.* **49**, 509–512 (1982).
13. D. R. Grempel, R. E. Prange, S. Fishman, *Phys. Rev. A Gen. Phys.* **29**, 1639–1647 (1984).
14. F. L. Moore, J. C. Robinson, C. F. Bharucha, B. Sundaram, M. G. Raizen, *Phys. Rev. Lett.* **75**, 4598–4601 (1995).
15. I. Manai *et al.*, *Phys. Rev. Lett.* **115**, 240603 (2015).
16. G. Lemarié *et al.*, *Phys. Rev. A* **80**, 043626 (2009).
17. M. Srednicki, *Phys. Rev. E Stat. Phys. Plasmas Fluids Relat. Interdiscip. Topics* **50**, 888–901 (1994).
18. M. Rigol, V. Dunjko, M. Olshanii, *Nature* **452**, 854–858 (2008).
19. S. Lellouch, A. Rançon, S. De Bièvre, D. Delande, J. C. Garreau, *Phys. Rev. A* **101**, 043624 (2020).
20. D. L. Shepelyansky, *Phys. Rev. Lett.* **70**, 1787–1790 (1993).
21. G. Gligorić, J. D. Bodyfelt, S. Flach, *Europhys. Lett.* **96**, 30004 (2011).
22. J. H. See Toh *et al.*, *Nat. Phys.* **18**, 1297–1301 (2022).
23. A. Cao *et al.*, *Nat. Phys.* **18**, 1302–1306 (2022).
24. C. Rylands, E. B. Rozenbaum, V. Galitski, R. Konik, *Phys. Rev. Lett.* **124**, 155302 (2020).
25. R. Chicireanu, A. Rançon, *Phys. Rev. A* **103**, 043314 (2021).
26. V. Vuatelet, A. Rançon, *Phys. Rev. A* **104**, 043302 (2021).
27. V. Vuatelet, A. Rançon, *Quantum* **7**, 917 (2023).
28. M. Fava, R. Fazio, A. Russomanno, *Phys. Rev. B* **101**, 064302 (2020).
29. B. Paredes *et al.*, *Nature* **429**, 277–281 (2004).
30. T. Kinoshita, T. Wenger, D. S. Weiss, *Science* **305**, 1125–1128 (2004).
31. E. Haller *et al.*, *Science* **325**, 1224–1227 (2009).
32. J. M. Wilson *et al.*, *Science* **367**, 1461–1464 (2020).
33. T. Kraemer *et al.*, *Appl. Phys. B* **79**, 1013–1019 (2004).
34. M. Olshanii, *Phys. Rev. Lett.* **81**, 938–941 (1998).
35. I. Bloch, J. Dalibard, W. Zwerger, *Rev. Mod. Phys.* **80**, 885–964 (2008).
36. T. Weber, J. Herbig, M. Mark, H.-C. Nägerl, R. Grimm, *Science* **299**, 232–235 (2003).
37. M. A. Cazalilla, R. Citro, T. Giamarchi, E. Orignac, M. Rigol, *Rev. Mod. Phys.* **83**, 1405–1466 (2011).
38. Y. Guo *et al.*, *Sci. Adv.* **10**, eadk6870 (2024).
39. Y. Guo *et al.*, *Nat. Phys.* **20**, 934–938 (2024).
40. See supplementary materials.
41. J. Lin, *IEEE Trans. Inf. Theory* **37**, 145–151 (1991).
42. M. Rigol, A. Muramatsu, *Phys. Rev. A* **72**, 013604 (2005).
43. M. Thamm, H. Radhakrishnan, H. Barghathi, B. Rosenow, A. Del Maestro, *Phys. Rev. B* **106**, 165116 (2022).
44. S. Tan, *Ann. Phys.* **323**, 2987–2990 (2008).
45. H. Yao, D. Clément, A. Minguzzi, P. Vignolo, L. Sanchez-Palencia, *Phys. Rev. Lett.* **121**, 220402 (2018).
46. C. Lim, K. Matirko, A. Polkovnikov, M. O. Flynn, arXiv:2401.01927 [cond-mat.stat-mech] (2024).
47. R. M. Nandkishore, S. L. Sondhi, *Phys. Rev. X* **7**, 041021 (2017).
48. N. Defenu *et al.*, *Rev. Mod. Phys.* **95**, 035002 (2023).
49. D. A. Abanin, E. Altman, I. Bloch, M. Serbyn, *Rev. Mod. Phys.* **91**, 021001 (2019).
50. M. Troyer, B. Ammon, E. Heeb, *Lect. Notes Comput. Sci.* **1505**, 191–198 (1998).
51. A. F. Albuquerque *et al.*, *J. Magn. Magn. Mater.* **310**, 1187–1193 (2007).
52. B. Bauer *et al.*, *J. Stat. Mech.* **2011**, P05001 (2011).
53. Y. Guo *et al.*, Observation of many-body dynamical localization [Data set], Zenodo (2024); https://doi.org/10.5281/zenodo.11229166.

ACKNOWLEDGMENTS

Funding: The Innsbruck team acknowledges funding by a Wittgenstein prize grant under Austrian Science Fund (FWF) project no. Z336-N36, by the European Research Council (ERC) under project no. 789017, by an FFG infrastructure grant with project no. FO999896041, and by the FWF's COE 1 and quantA. M.H. thanks the doctoral school ALM for hospitality, with funding from the FWF under project no. W1259-N27. The team at Zhejiang University is supported by the National Key R&D Program of China (grant no. 2022YFA1404203) and the National Natural Science Foundation of China (grant no. 12375021). H.Y. is also supported by the Swiss National Science Foundation under grant no. 200020-188687. **Author contributions:** The work was conceived by H.-C.N., Y.G., M.L., L.Y., S.D., and Z.C. Experiments were performed by Y.G. and S.D. Data were analyzed by Y.G., A.Y., and Z.C. Theoretical models and simulation were done by L.Y., A.Y., H.Y., and Z.C. Preparation of experiments was done by Y.G, S.D., and M.H. Manuscript preparation was mainly contributed by H.-C.N., Y.G., L.Y., Z.C., A.Y., and M.L. **Competing interests:** The authors declare no competing interests. **Data and materials availability:** The quantum Monte Carlo calculations make use of the ALPS scheduler library and statistical analysis tools (*50*–*52*). The data shown in the main text and supplementary materials are available at Zenodo (*53*).

License information: Copyright © 2025 the authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original US government works. https://www.science.org/about/science-licenses-journal-article-reuse

SUPPLEMENTARY MATERIALS

science.org/doi/10.1126/science.adn8625
Materials and Methods; Supplementary Text; Figs. S1 to S9; References (*54*–*59*)

Submitted 4 January 2024; accepted 17 June 2025

10.1126/science.adn8625

Downloaded from https://www.science.org at Hong Kong Polytechnic University on March 16, 2026

Observation of many-body dynamical localization

Yanliang Guo, Sudipta Dhar, Ang Yang, Zekai Chen, Hepeng Yao, Milena Horvath, Lei Ying, Manuele Landini, and Hanns-Christoph Nägerl

Science **389** (6761), . DOI: 10.1126/science.adn8625

Editor's summary

Interactions in a many-body system often lead to thermalization. However, it has recently been theoretically predicted that an interacting one-dimensional bosonic gas subject to periodic "kicks" localizes instead of thermalizing. Guo *et al*. realized such a system using quantum-degenerate samples of cesium atoms and tuned the interaction strength from noninteracting to strongly interacting, finding localization in both limits. —Jelena Stajic

View the article online

https://www.science.org/doi/10.1126/science.adn8625

Permissions

https://www.science.org/help/reprints-and-permissions

Use of this article is subject to the Terms of service

Science (ISSN 1095-9203) is published by the American Association for the Advancement of Science. 1200 New York Avenue NW, Washington, DC 20005. The title *Science* is a registered trademark of AAAS.

Copyright © 2025 The Authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original U.S. Government Works

Downloaded from https://www.science.org at Hong Kong Polytechnic University on March 16, 2026